FLD 301 · THE FIELD TERM · FIELD REPORT

Twelve Weeks at an Instrument Shop

Rowan Hale · placement: Sea Grant instrument shop, Point Loma · mentor: J. Alvarez · 8 Jan – 30 Mar 2030

The shop builds and repairs the instruments the oceanographers put in the water: CTDs, current meters, the cabled sensors on the pier. I went to learn how instruments survive the sea. I came back having learned mostly about the people who keep checking them.

Summary. Twelve weeks, four days a week. Weeks 1–3 on the bench, rebuilding connectors and potting cables. Weeks 4–8 on a mooring refit, where I built the housing that became ENGR 320's logger. Weeks 9–12 on my own project with the shop's blessing: a maintenance log for the pier's instruments, which they still use.

What I assumed. That saltwater is the enemy and the work is sealing things against it.

What I found. Salt is slow. Chafe, fatigue, and forgetting are fast. Nine of the eleven failures I saw in twelve weeks were mechanical, at a point where something moved against something else, or they were human: a thing that had worked for so long that nobody opened it. The instruments that lasted were the ones with a date written on them.

Weeks 1–6

Week 1. Twenty-two underwater connectors, cleaned and re-greased. J. made me do the first one three times. By the twentieth I understood why: the grease is not the seal. The clean surface under it is.

Week 2. Potting cables in urethane. Bubbles are failures you cannot see for six months. We learned to pour slowly, in a warm room, and to write the pour date on the boot.

Week 3. A current meter back from eighteen months at 300 m, perfect inside, a bearing ground to dust outside. The first time I understood that the sea's main tool is repetition.

Week 4. The mooring refit begins. I assumed the salt was the enemy. It was the cable chafing on the piling. I noticed in week four, on a cable that had been chafing for a year.

Week 5. Designed the housing for my own logger from what the shop's housings taught me: a straight pull on every cable, no sharp bends, a bracket that can be inspected without a diver.

Week 6. First deployment of the logger from the shop's skiff. It came back dry. J. said: good, now do it ten more times before you believe it.

THE LOG, AND WHAT IT CHANGED

Weeks 7–12

Week 7–8. Mooring back in the water. I kept a notebook of every assumption I made and every one that turned out wrong. Eleven of thirty-four. The wrong ones were all of the same kind: I assumed something that had not failed would not fail.

Weeks 9–12. The shop had no single place where the pier's instruments' service history lived. I built one: a plain spreadsheet and a laminated card by the door, with the date each thing was last opened. It is not technology. It is a habit with a surface. They still use it, and the bracket on my own logger is on it, with the date I should have checked it and did not.

What I'd tell the next learner. Go to the bench first and the design second. Write the date on everything. Ask what failed last year before you ask what the sensor measures.

J. ALVAREZ · SHOP LEAD · EXCERPT, WITH PERMISSION

Mentor evaluation

"Rowan arrived wanting to design and left wanting to maintain, which is the right direction. Careful hands by week three. Asks the right question, which is 'what broke last time.' The maintenance card is the most useful thing a student has left here in years because it is the simplest. Would take them back. Would hire them after the degree."

Hours: 384. Attendance: full. Unassisted assessment (the shop's own, no notes): rebuild a connector and explain each step. Passed.

Reviewed against the rubric for the Field Term: Judgment, proficient. The report, the journal (12 weeks, 41 pages), and the evaluation enter the Portfolio together.